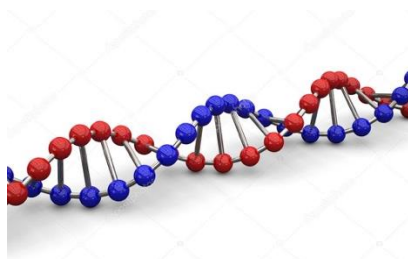
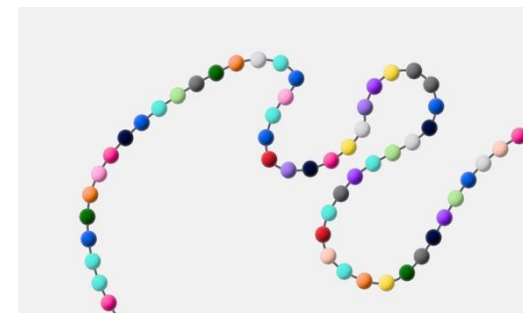




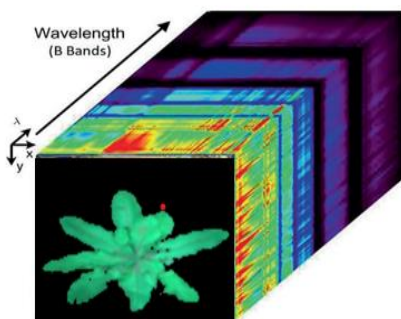
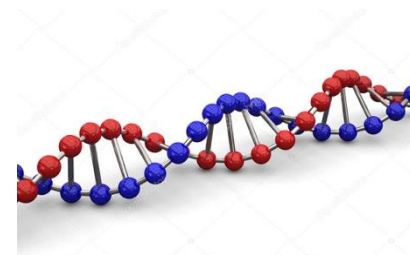
Foundation models and self-supervised learning



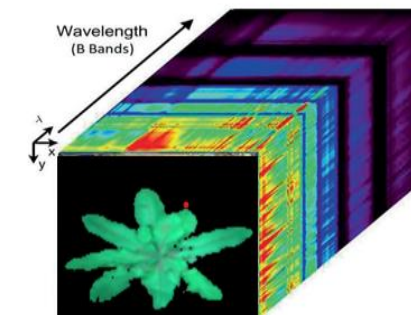
**Protein foundation
model**



**DNA foundation
model**



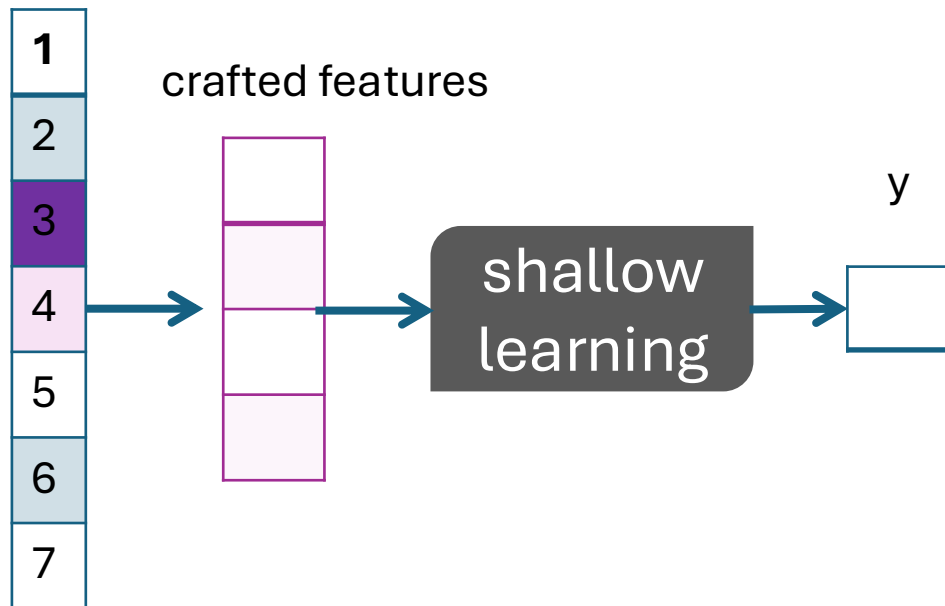
**Image foundation
model**



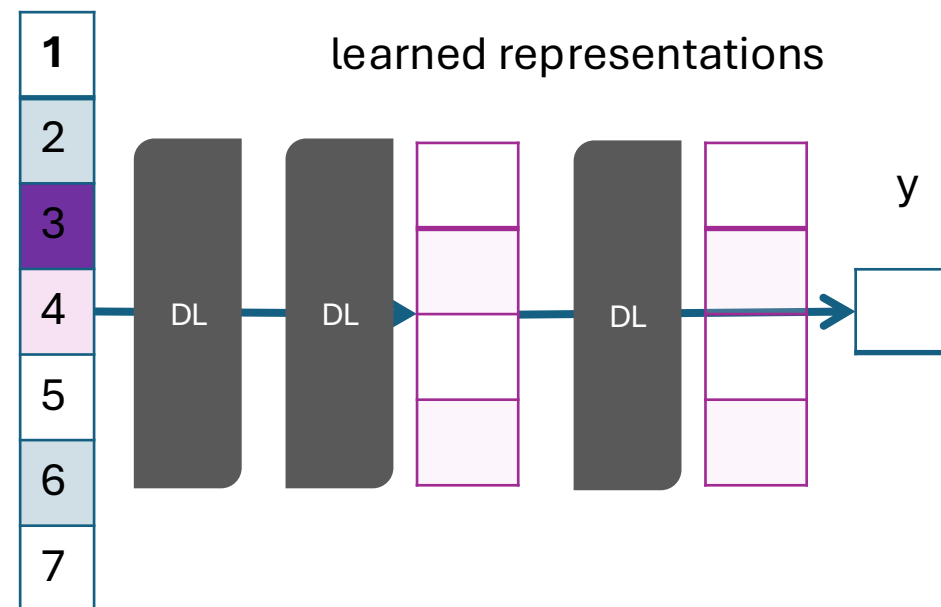


Deep learning vs shallow learning

sequence



sequence



Deep learning:

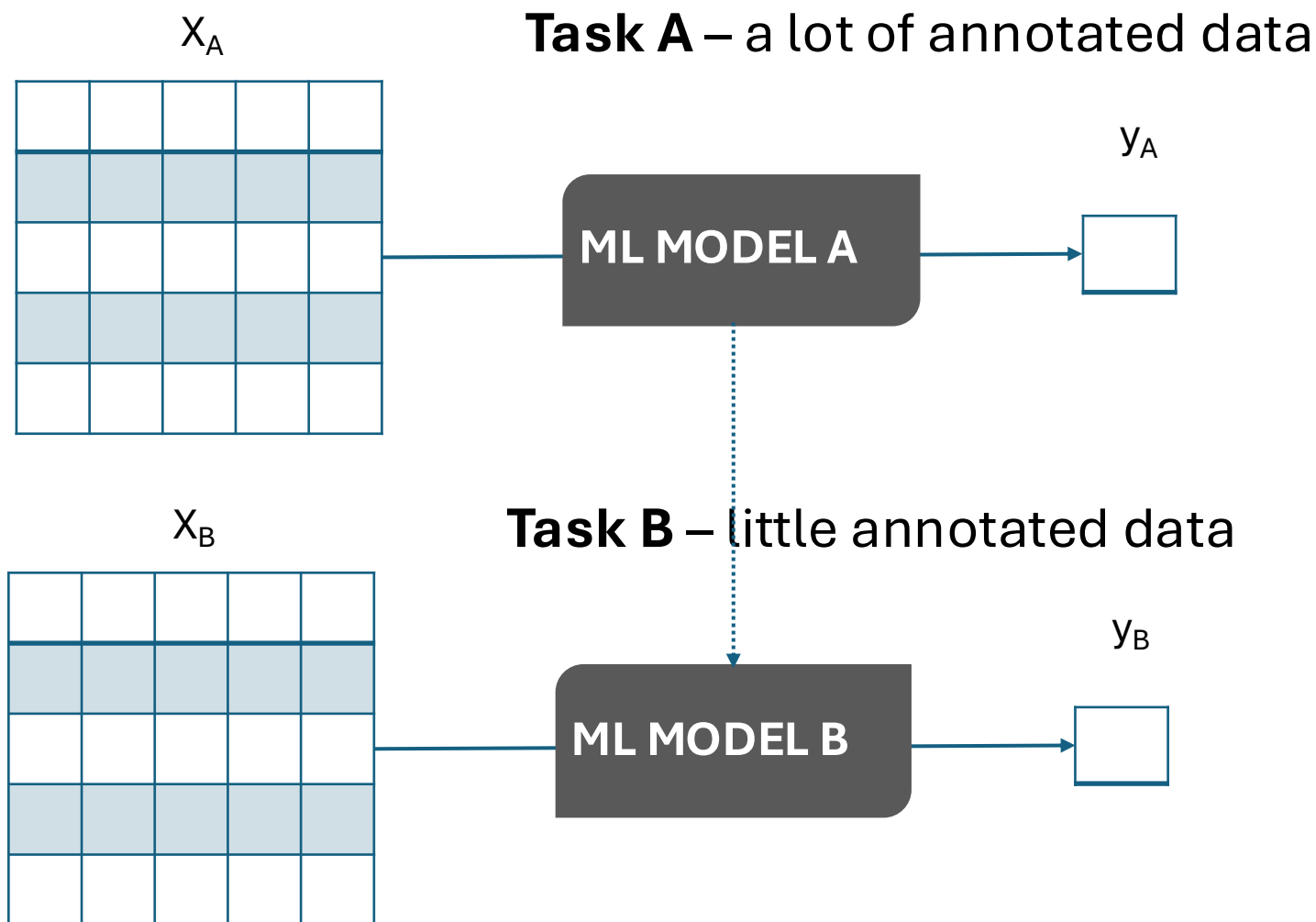
End-to-end learning; raw input data is used, representations using the data alone (500-1000 examples per class)

Shallow learning:

Hand-crafted features are derived from the sequence; requires limited annotations (50 examples per class)



Transfer learning

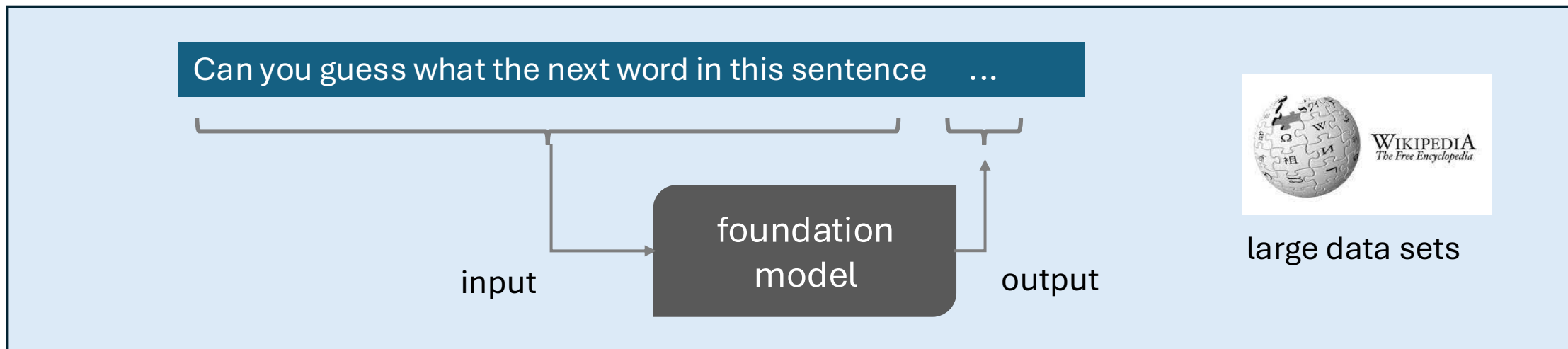


- Can we use a Model A, trained on a similar task, with a lot of available data, for a task B, with little annotated data.
- How would you do this?

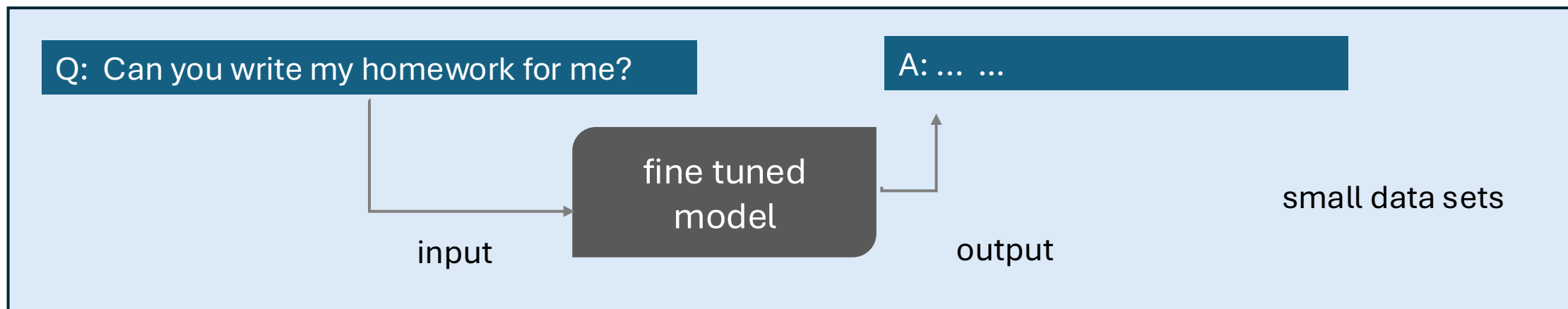


Foundation model (for specific data type / modality)

Training



fine tuning

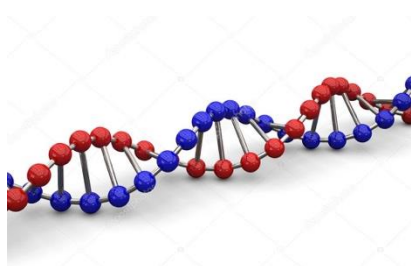
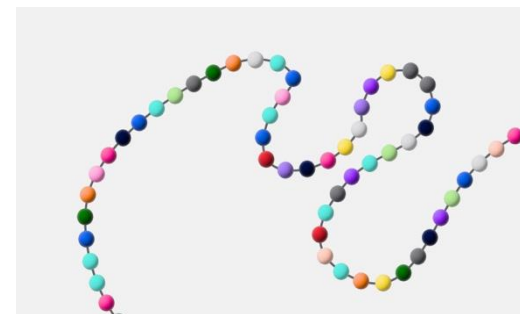




Foundation models in the Life Sciences



Protein foundation
model



DNA foundation
model

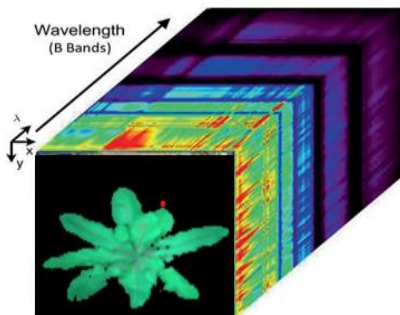
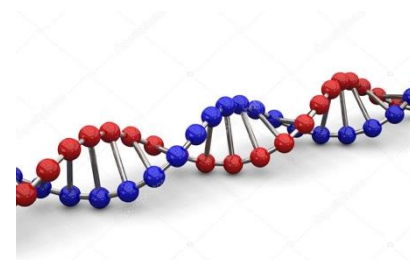
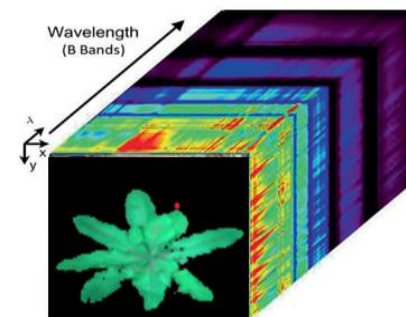
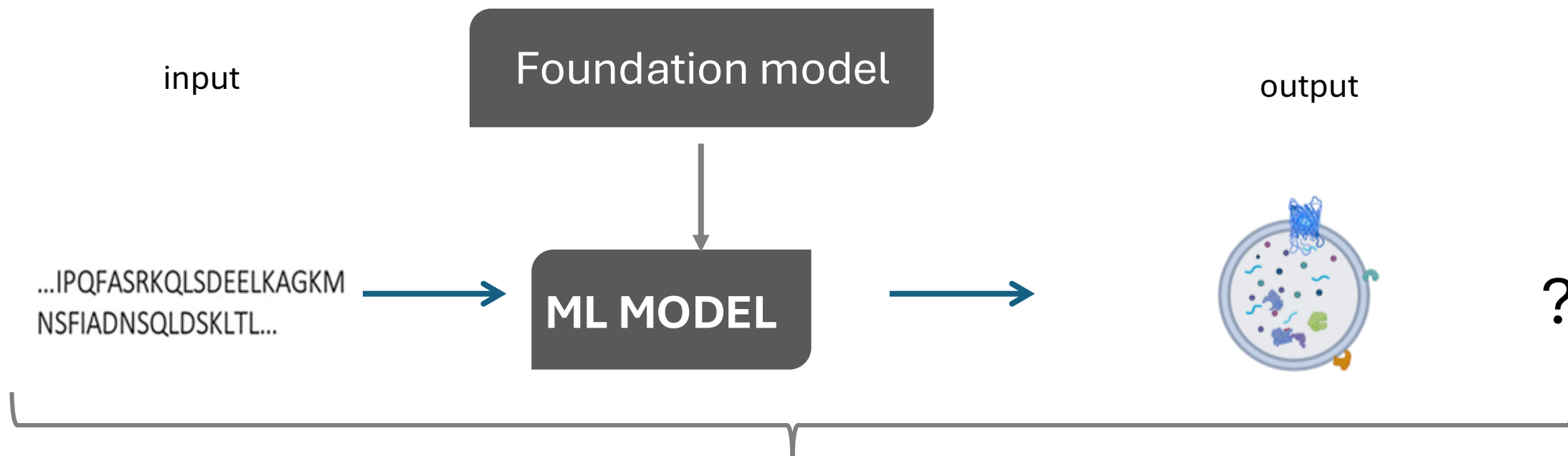


Image foundation
model





Fine tuning for down stream tasks



Down stream task:

- Extra cellular vesicle prediction from protein sequence
- Using a sequence-based protein foundation model for representation learning.



Down stream tasks

Protein language model

DNA language model

Image feature extractor /
representation learning



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Enter the event code in the top banner

Event code

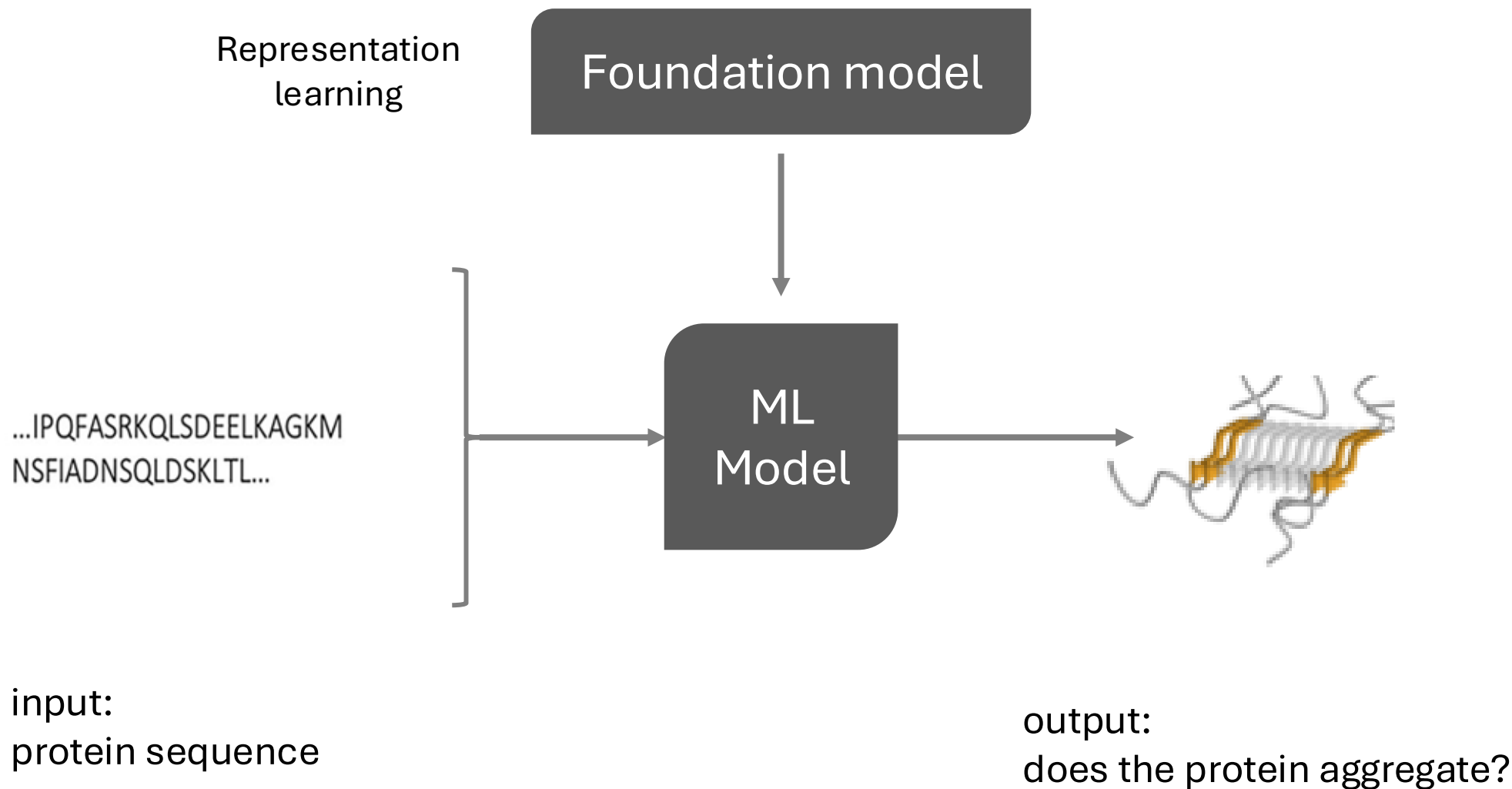
EQRXHUE



Waiting for participants ...

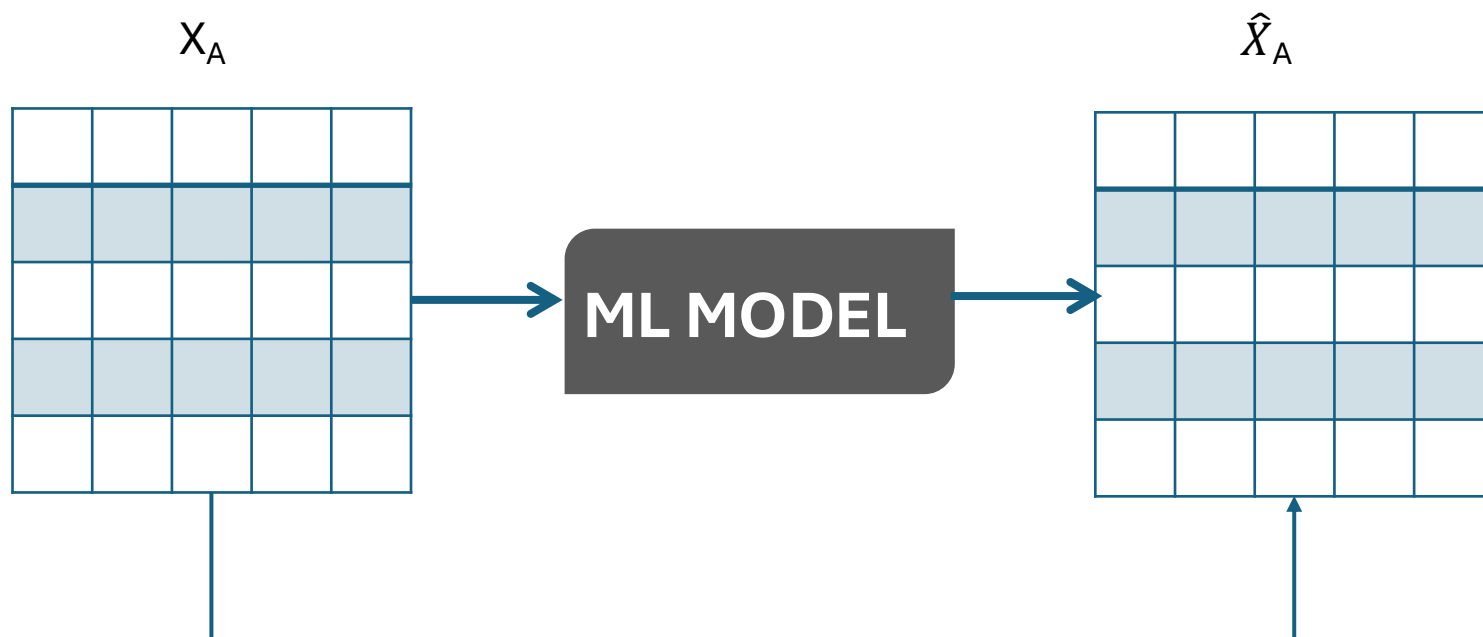


Fine tuning Foundation Models (Assignment 2)





Self supervised learning



Data reconstruction

Self-supervised learning task:

- Learn to reconstruct the input data

Why?

- To learn representations of the data, without labelled annotations
- To extract features
- For “down-stream” task

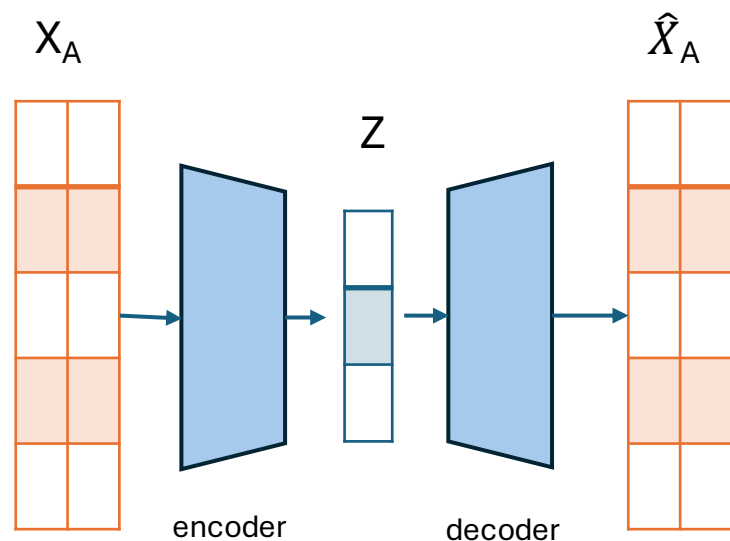
But how will it learn?

- What would be a trivial data transformation to reconstruct the data?



Self-supervised learning: two main learning strategies

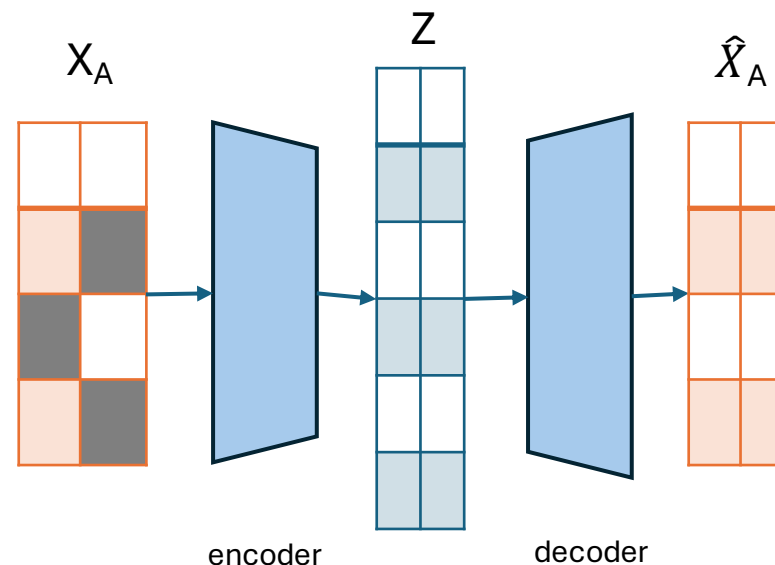
Bottleneck



The latent representation (Z), has a lower dimensionality than the input (and output data).

This enforces the model to “compress” the information, and thereby to learn a more effective representation.

Masking

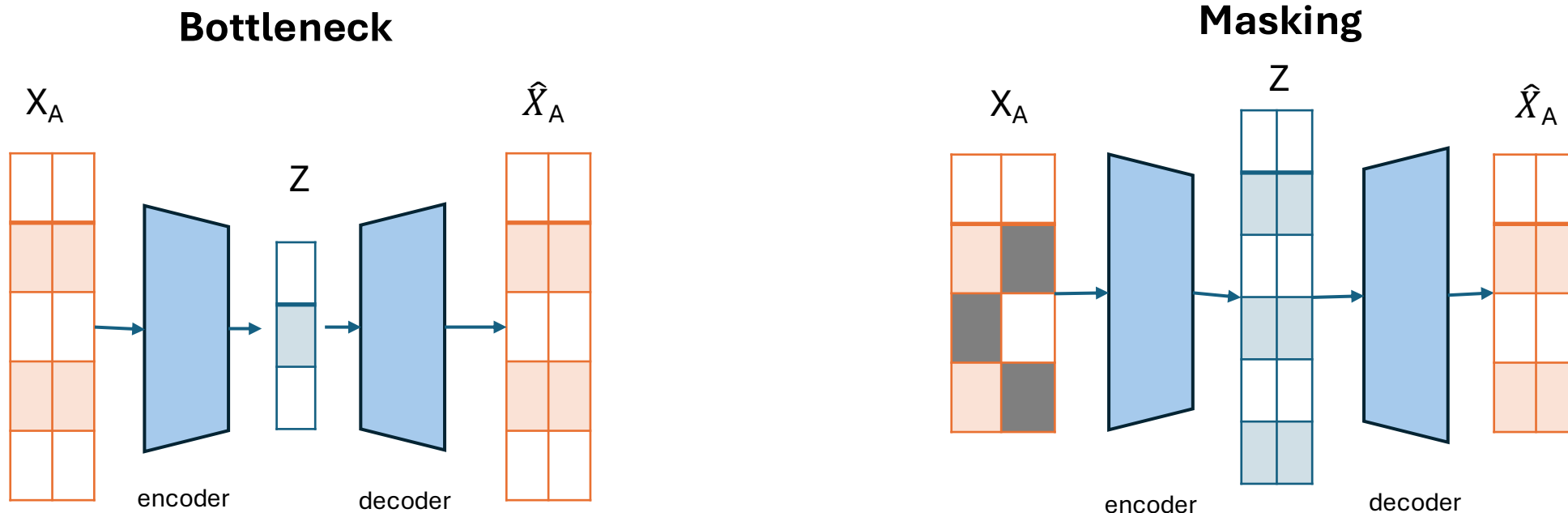


There is no constraint on dimensionality of Z with respect to the input (and output data). In fact, the dimensionality may be larger for Z than for X .

Part of the input data is masked: the model needs to learn to reconstruct the data.



Self-supervised learning: two main strategies



Both of these approaches are classed as “generative AI”: the models can generate new data.

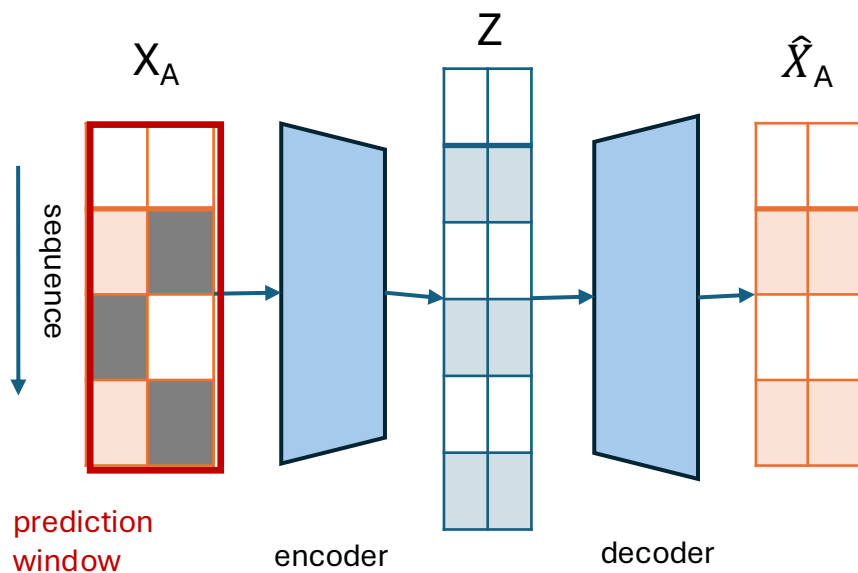
Auto-encoders and Variational auto-encoders are well known examples to the bottleneck strategy.

Most large language models and foundation models use a masking approach.

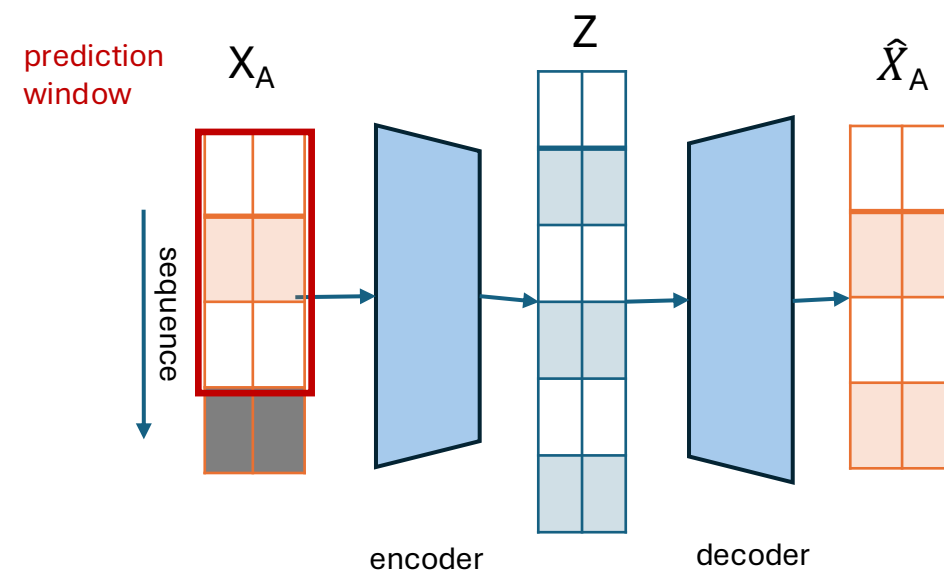


Masking strategies

Bert-like-objective

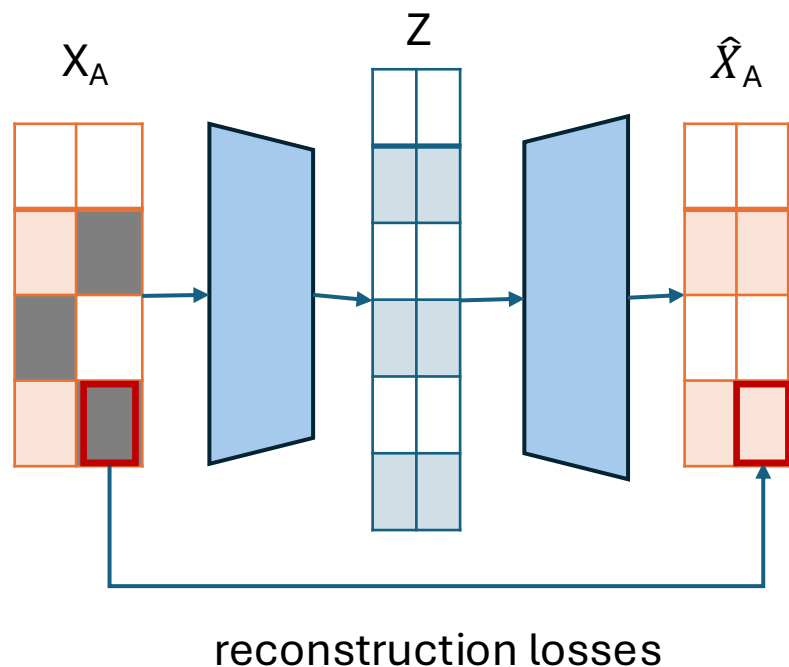


Next in sequence objective





Reconstruction loss



Regression losses (images)

Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{x}_i|$$

Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

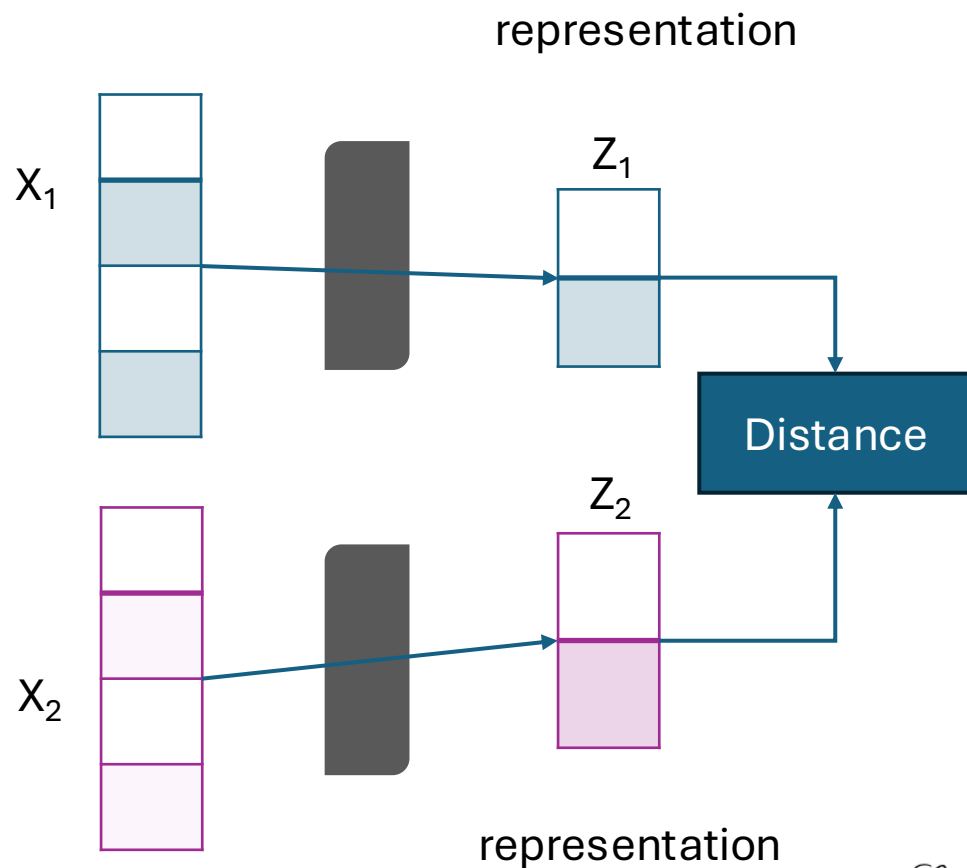
Classification losses (protein sequences, text)

- Cross entropy (non-binary)

$$\text{CE} = -\frac{1}{n} \sum_{i=1}^n \sum_{c=1}^C P(x_i = c) \log P(\hat{x}_i = c | x_i)$$



Contrastive loss



A contrastive loss can for example be used to align representations of different modalities

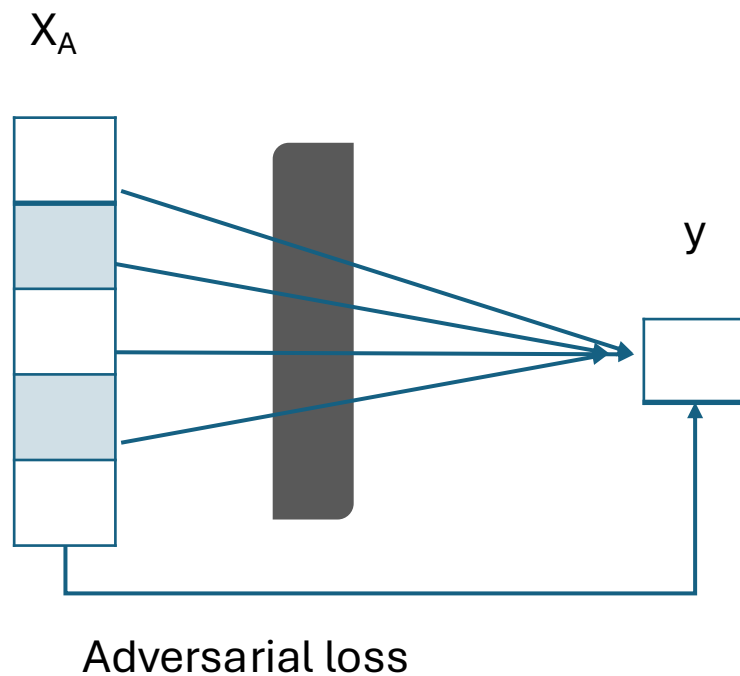
same class label
(positive pair) $c_{ij}=1$

different class label
(negative pair) $c_{ij}=0$

$$\mathcal{L} = \frac{1}{2n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n [c_{ij} \cdot D(z_i, z_j)^2 + (1 - c_{ij}) \cdot \max(0, m - D(z_i, z_j))^2]$$



Adversarial loss



- A loss pointing the “wrong way”
- You are optimising weight such that the model cannot classify
- How may this be helpful?



Transformers and attention

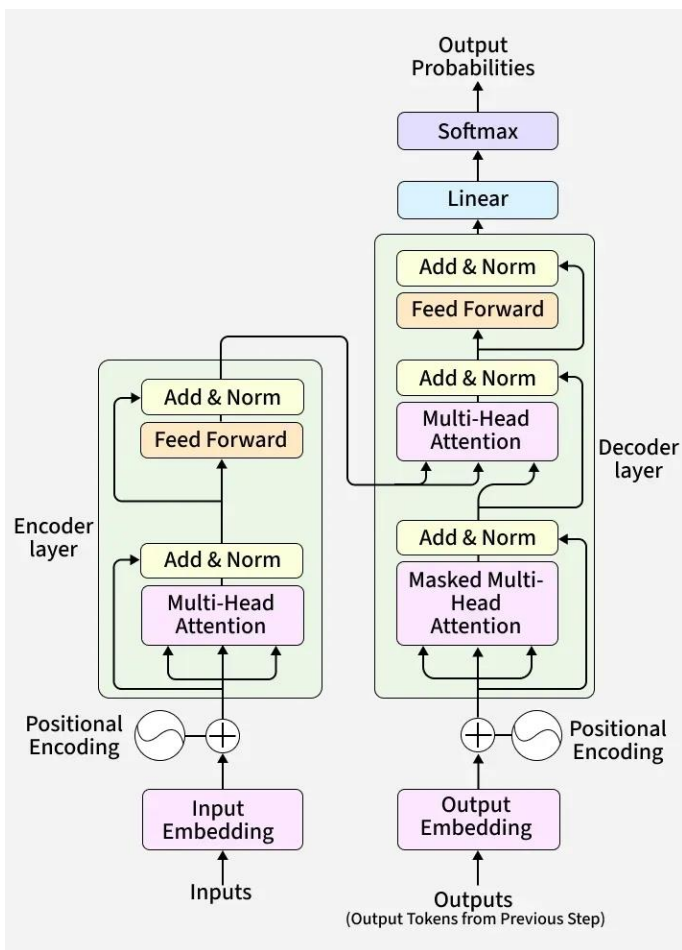
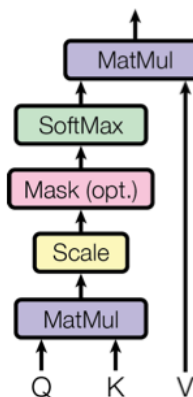
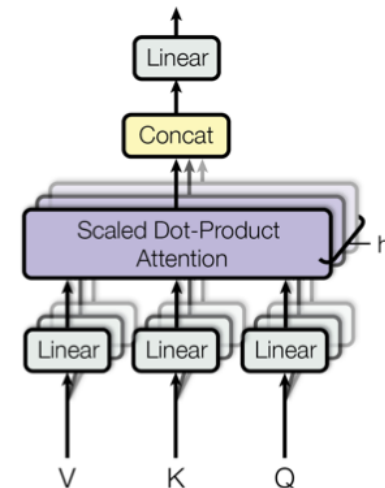


Figure 1: The Transformer - model architecture.

Scaled Dot-Product Attention



Multi-Head Attention



Nomenclature from language processing

Q: query

K: key

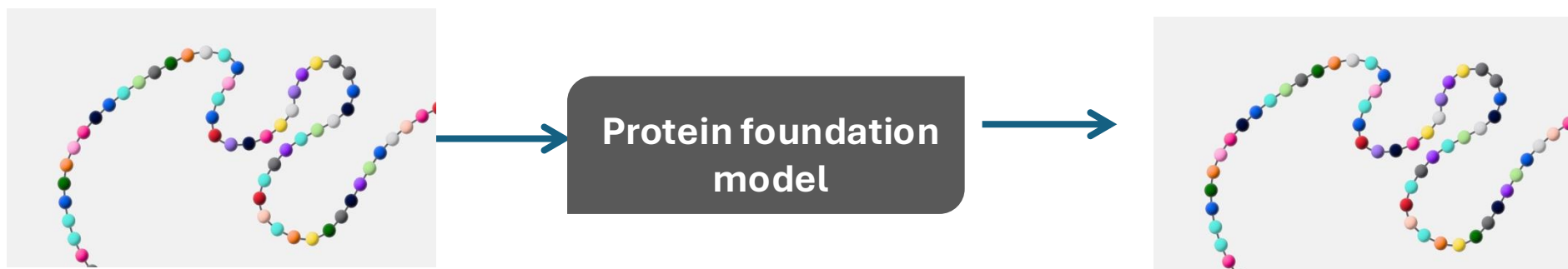
V: value

Idea: give context dependent attention.

Weights learn on what data to point the attention



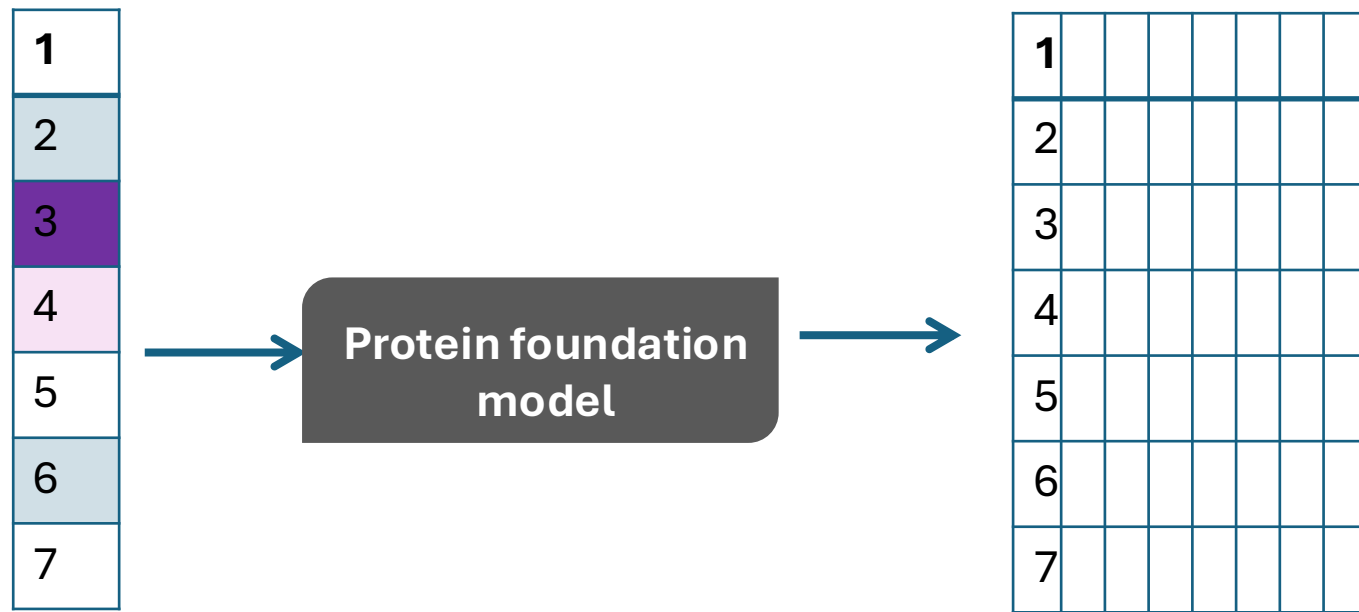
Protein language models



Protein language models, are trained in a self-supervised manner, over > billion protein sequences.



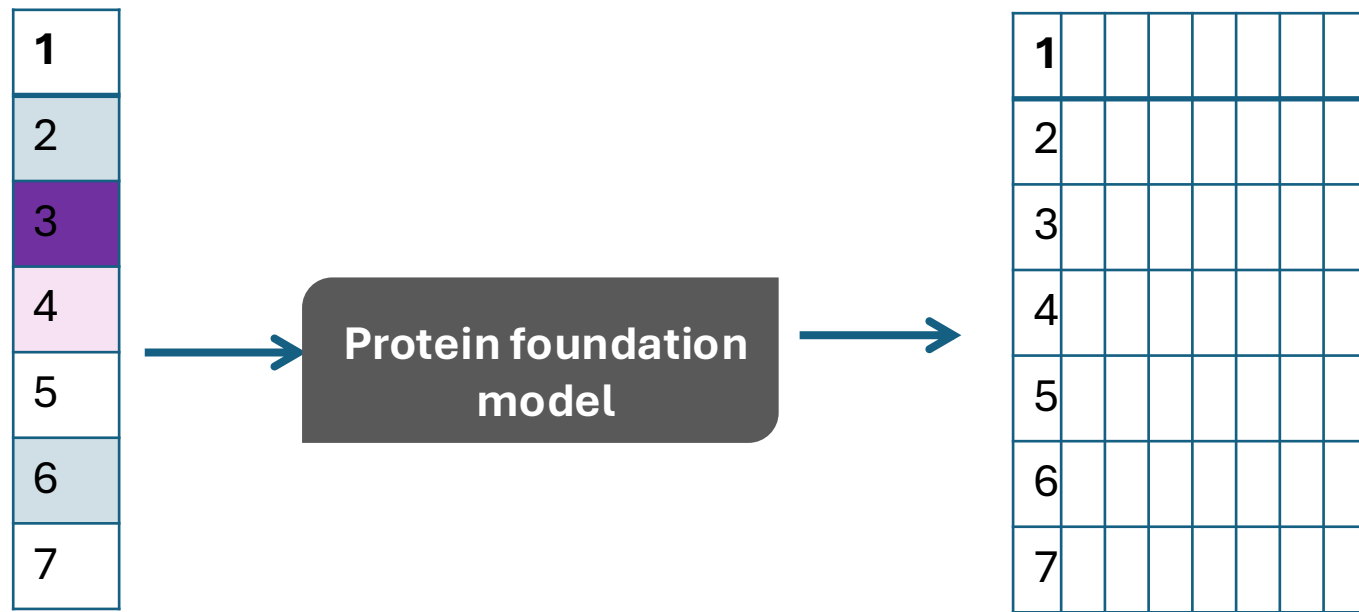
Protein language models



- Protein language models can provide embedding for amino acid sequences.
- They provide a context, in terms of a vector per sequence position.
- It increases the dimensionality of your representation



Protein language models



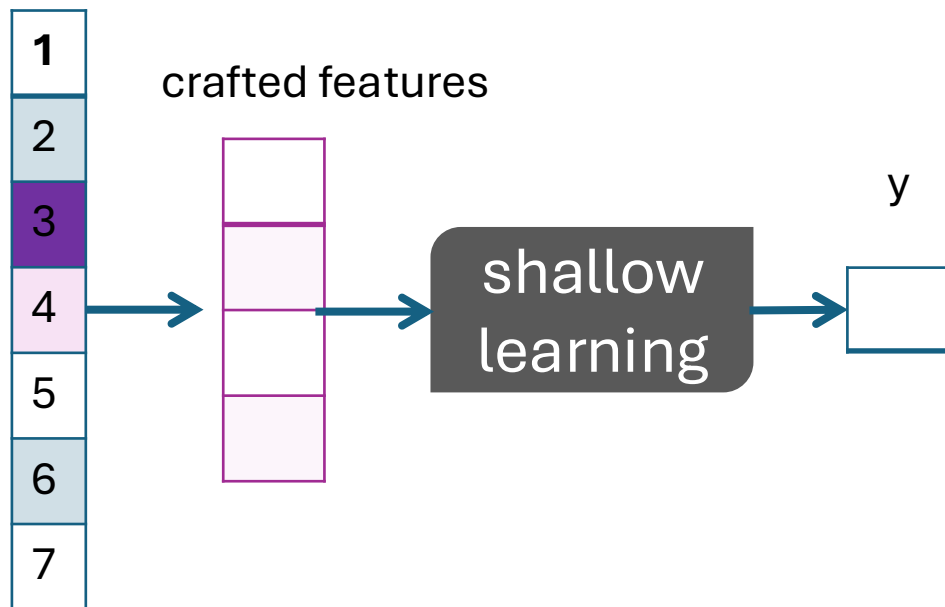
Question?

How could increasing the dimensionality of your problem, help with downstream tasks for which little annotations are available?

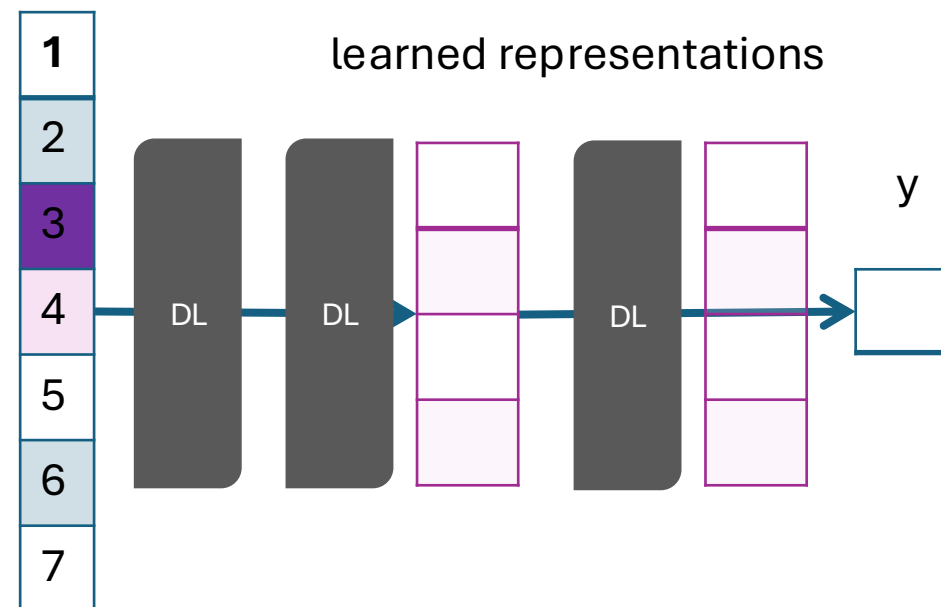


Deep learning vs shallow learning

sequence



sequence



Deep learning:

End-to-end learning; raw input data is used, representations using the data alone (500-1000 examples per class)

Shallow learning:

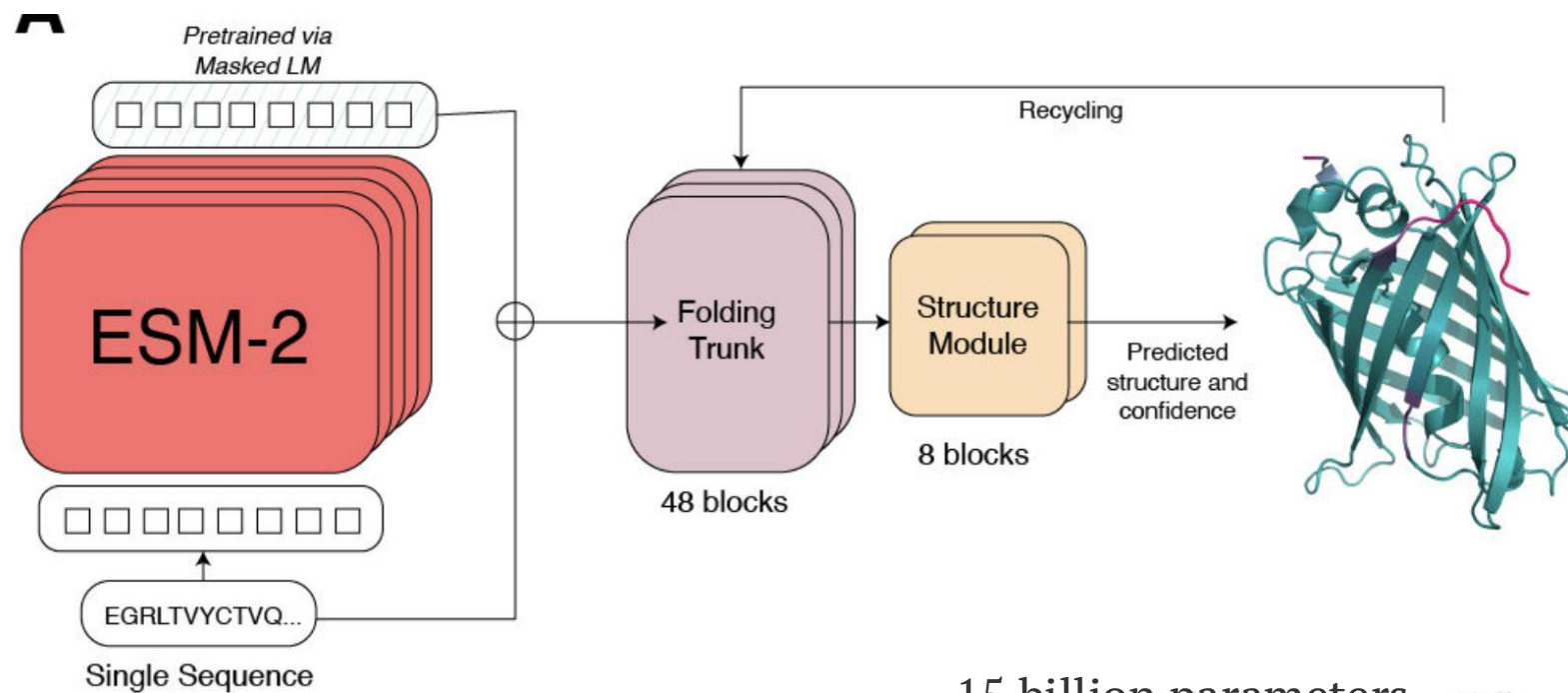
Hand-crafted features are derived from the sequence; requires limited annotations (50 examples per class)



ESM

Protein language model

Down stream task



15 billion parameters - ...

600 million metagenomic proteins



CARP – a convolution based foundation model

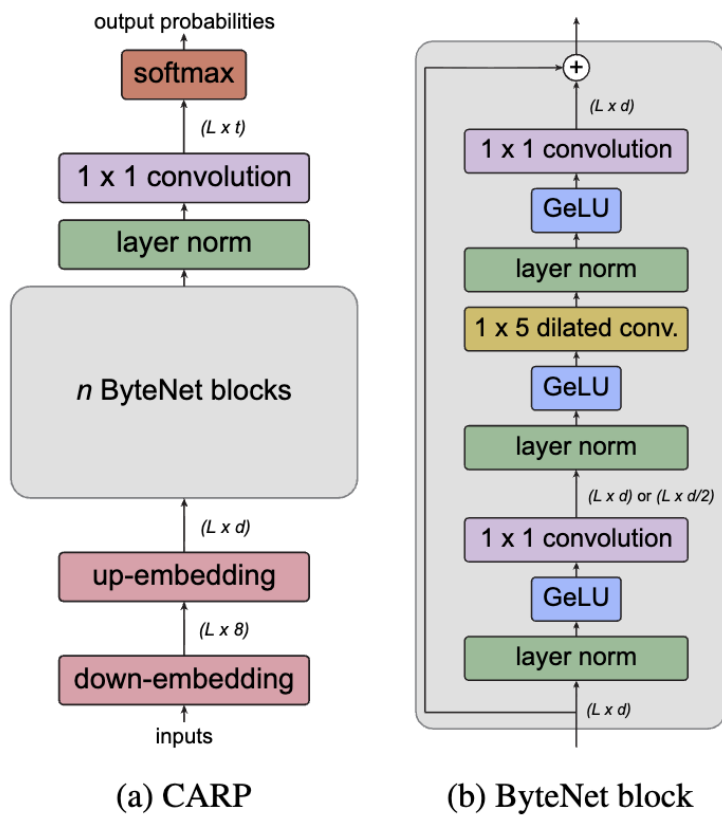
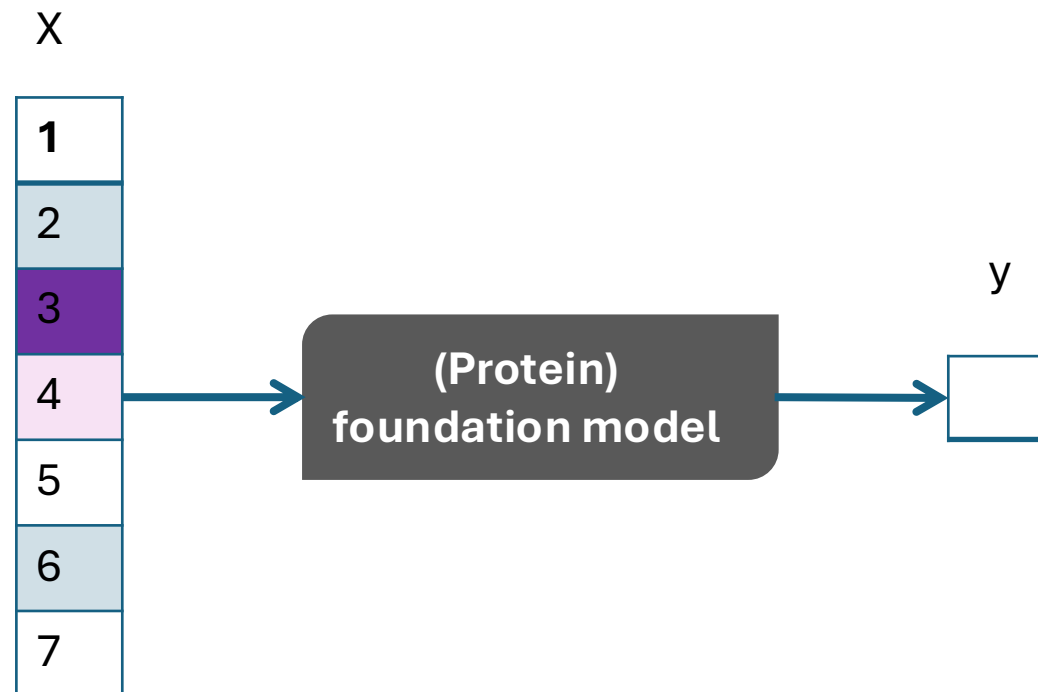


Figure 1: The CARP architecture.

- Convolution based foundation models
- Can also provide embeddings for protein sequence
- Much faster to train, and to fine-tune.



Few shot learning



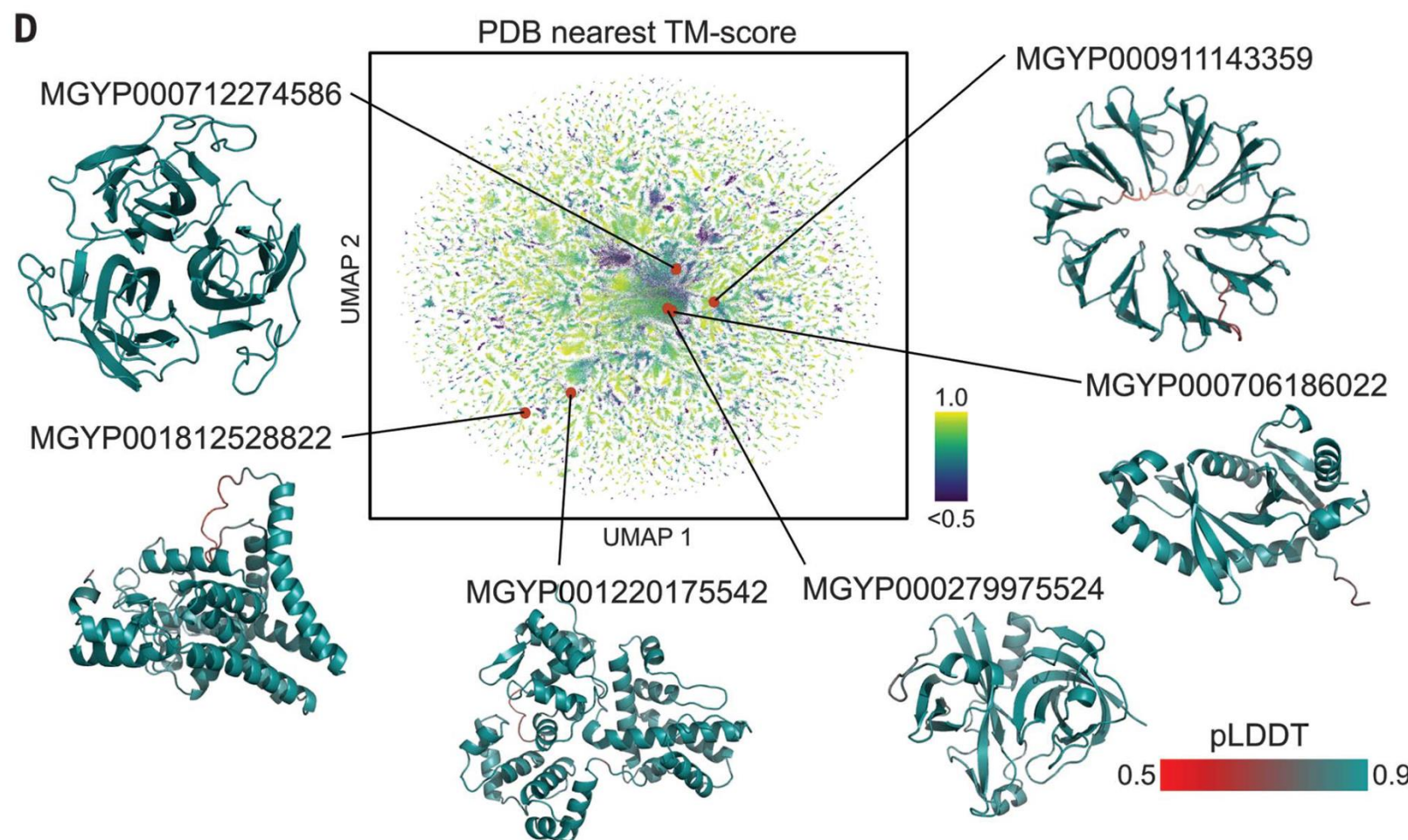
- Training a problem on a very limited annotated dataset (5-20 samples)
- Essentially, the model only learns to map the embeddings of the foundation model, to the class labels



Zero shot learning

- Finding novel classes in data, unseen in original training data set. E.g. finding a new protein family.
- Distances between embedding can for example also be used for impact prediction of a specific mutation (a change in amino acid)

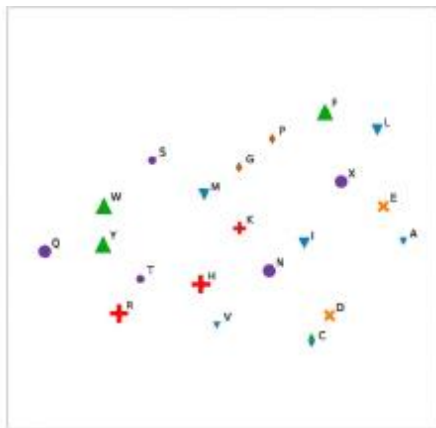
UMAP over ESM2 embeddings



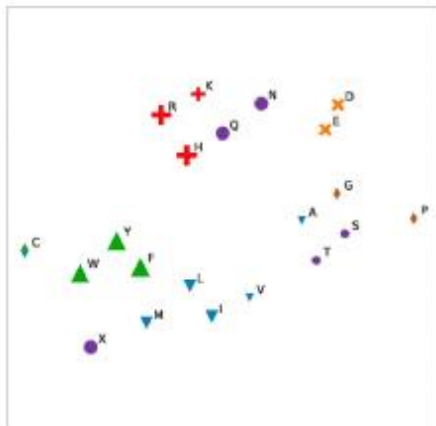


What Information Exists Inside Embeddings?

Random

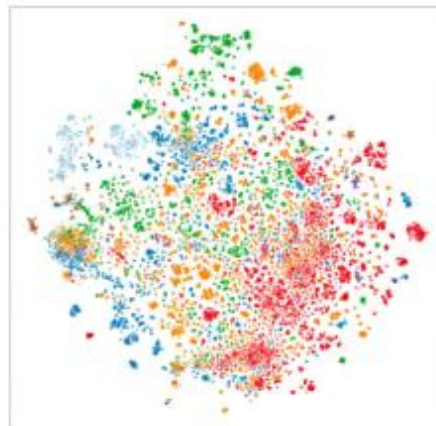
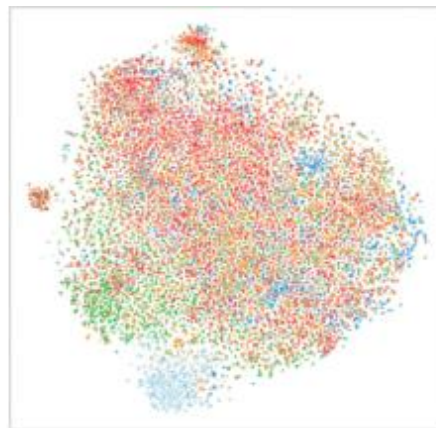


Pre-trained



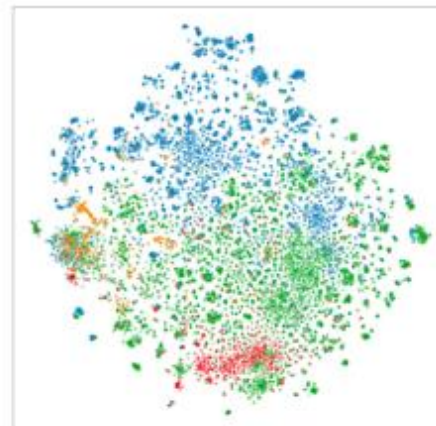
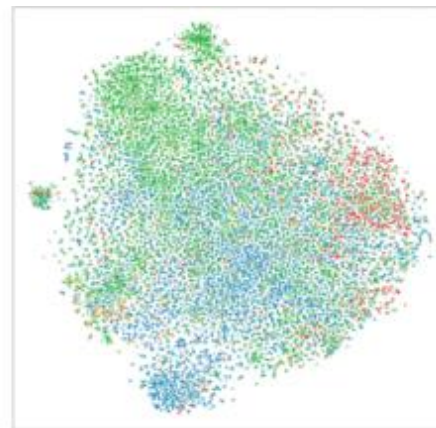
- ▲ Hydrophobic (aromatic) ◆ Special cases
- ▼ Hydrophobic (aliphatic) ● Small (<130 Dalton)
- + Positive ● Medium
- × Negative ● Big (>150 Dalton)
- Polar neutral

A Amino acids



- All alpha
- All beta
- Alpha & beta (a|b)
- Alpha & beta (a+b)
- Multi-domain
- Membrane, cell surface
- Small proteins

B Structure: SCOPe

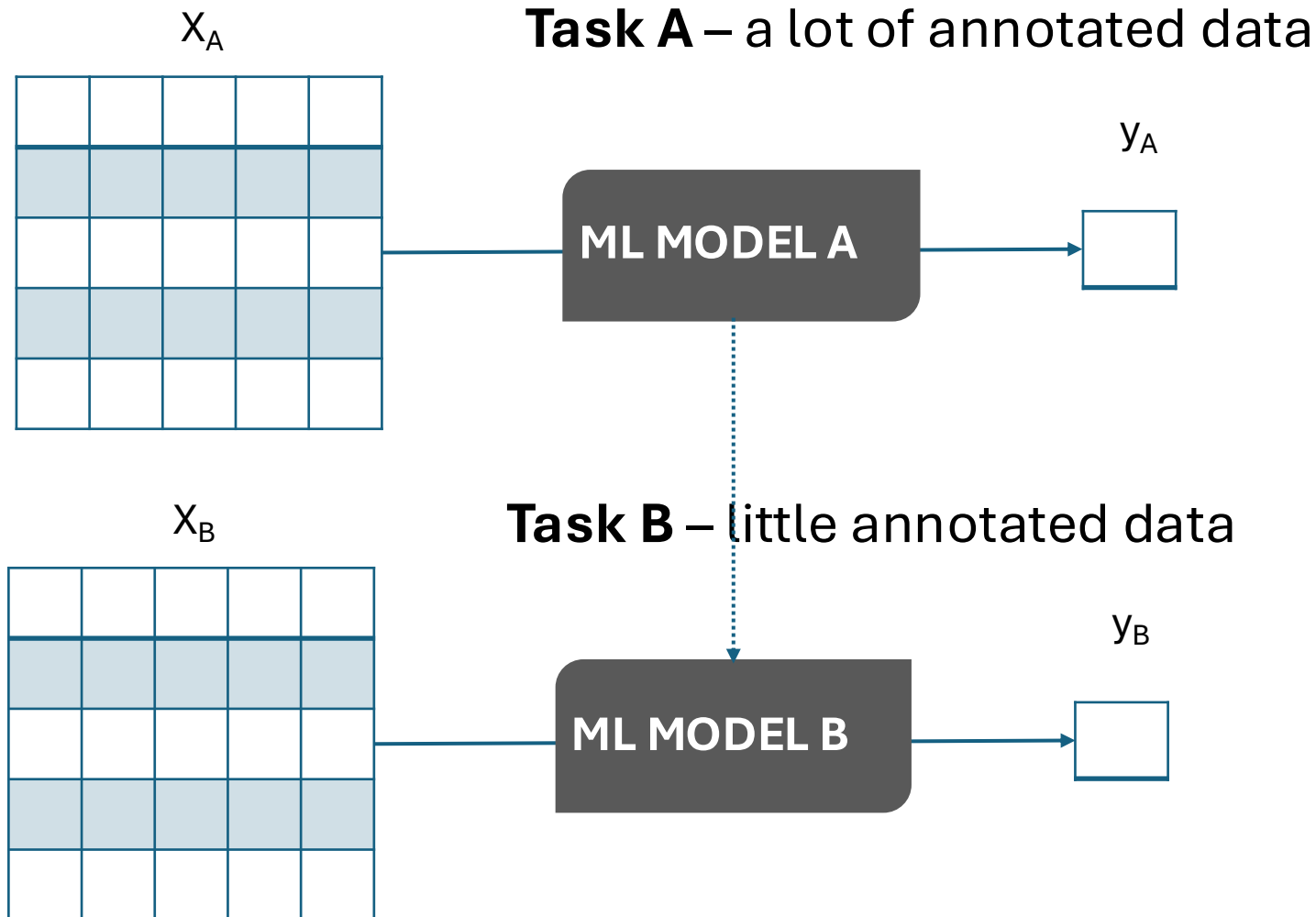


- Eukaryota
- Bacteria
- Archaea
- Viruses

C Lineage: Kingdoms



Transfer learning

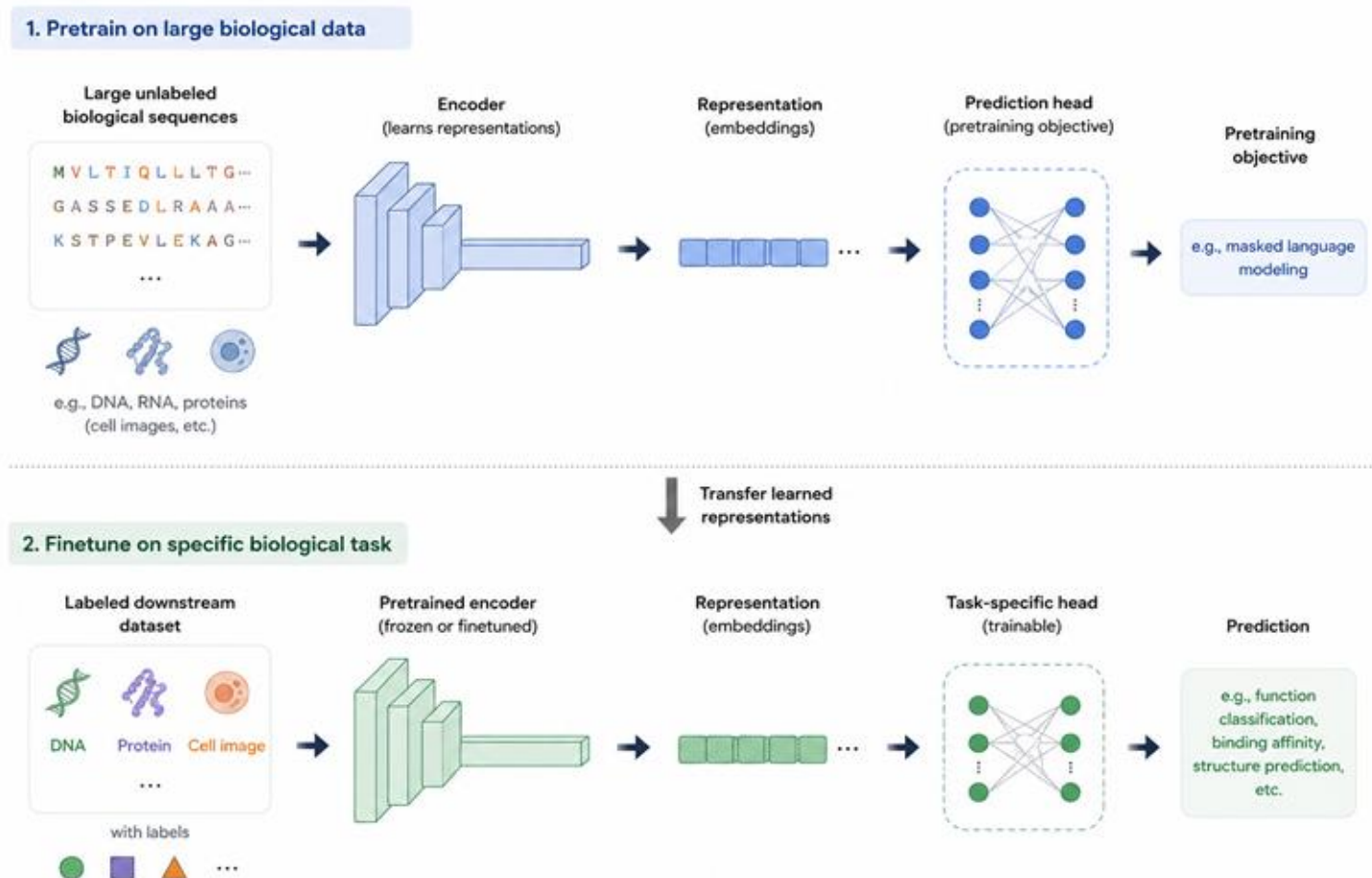


- Can we use a Model A, trained on a similar task, with a lot of available data, for a task B, with little annotated data.
- How would you do this?



Transfer Learning and foundation model fin-tuning

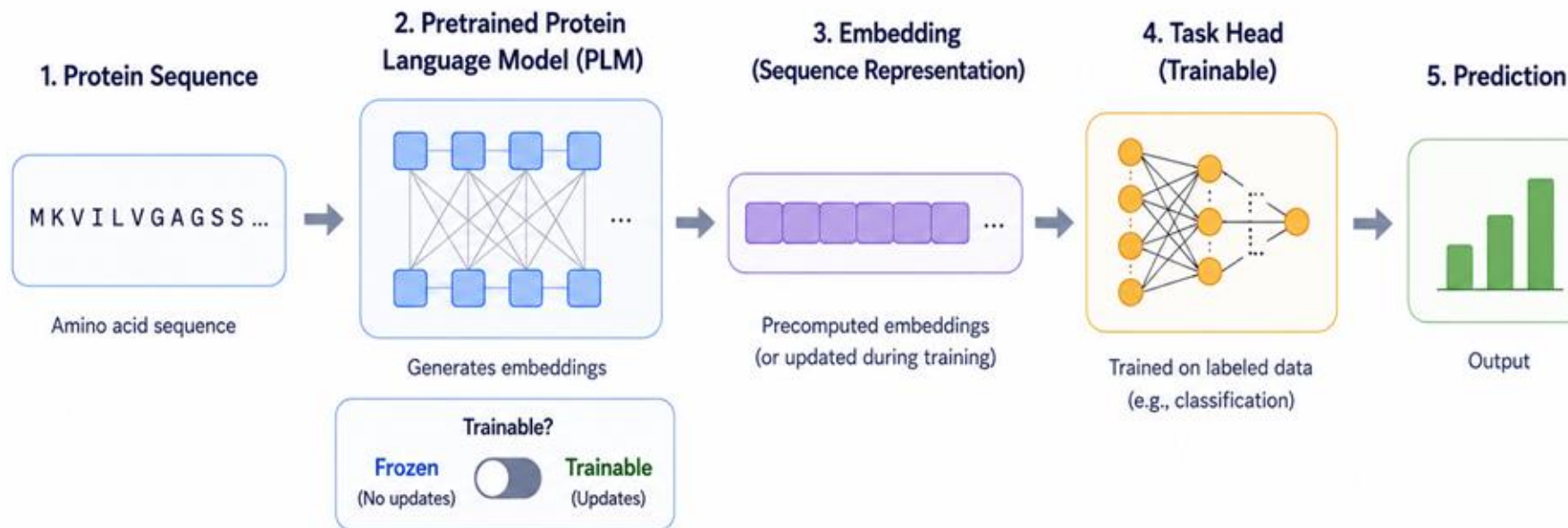
- Pretrain once on massive datasets
- Reuse model for smaller tasks
- Reduces need for expensive labels
- Improves performance on limited datasets





Frozen Fine Tuning vs Full Fine Tuning

- Extract embeddings
- Train small classifier/regressor
- Encoder weights of foundation model remain fixed
- Update foundation model weights





Fine-tuning protein language models boosts predictions across diverse tasks

Received: 25 January 2024

Accepted: 15 August 2024

Published online: 28 August 2024

 Check for updates

Robert Schmirler ^{1,2,3}✉, Michael Heinzinger ¹ & Burkhard Rost ^{1,4,5}

Prediction methods inputting embeddings from protein language models have reached or even surpassed state-of-the-art performance on many protein prediction tasks. In natural language processing fine-tuning large language models has become the de facto standard. In contrast, most protein language model-based protein predictions do not back-propagate to the language model. Here, we compare the fine-tuning of three state-of-the-art models (ESM2, ProtT5, Ankh) on eight different tasks. Two results stand out. Firstly, task-specific supervised fine-tuning almost always improves downstream predictions. Secondly, parameter-efficient fine-tuning can reach similar improvements consuming substantially fewer resources at up to 4.5-fold acceleration of training over fine-tuning full models. Our results suggest to always try fine-tuning, in particular for problems with small datasets, such as for fitness landscape predictions of a single protein. For ease of adaptability, we provide easy-to-use notebooks to fine-tune all models used during this work for per-protein (pooling) and per-residue prediction tasks.



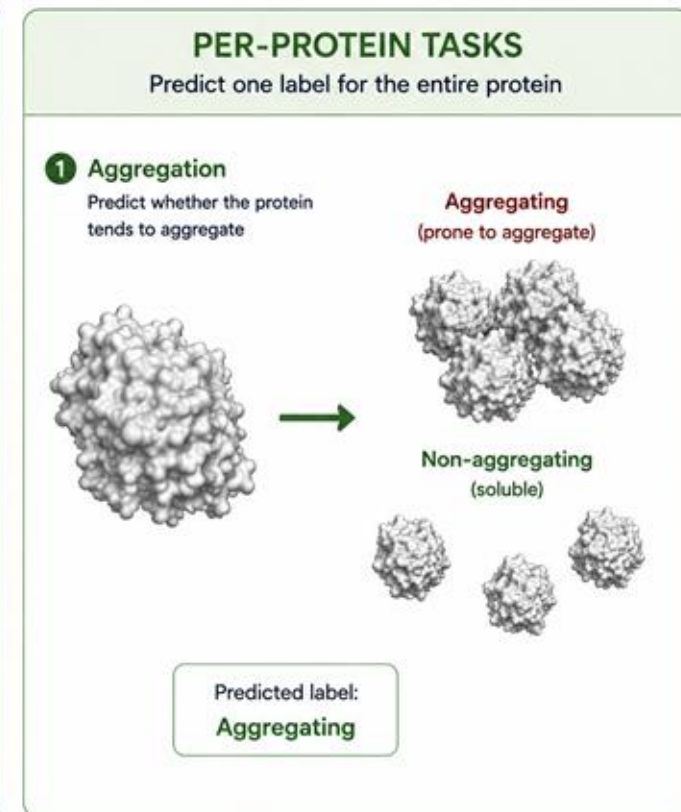
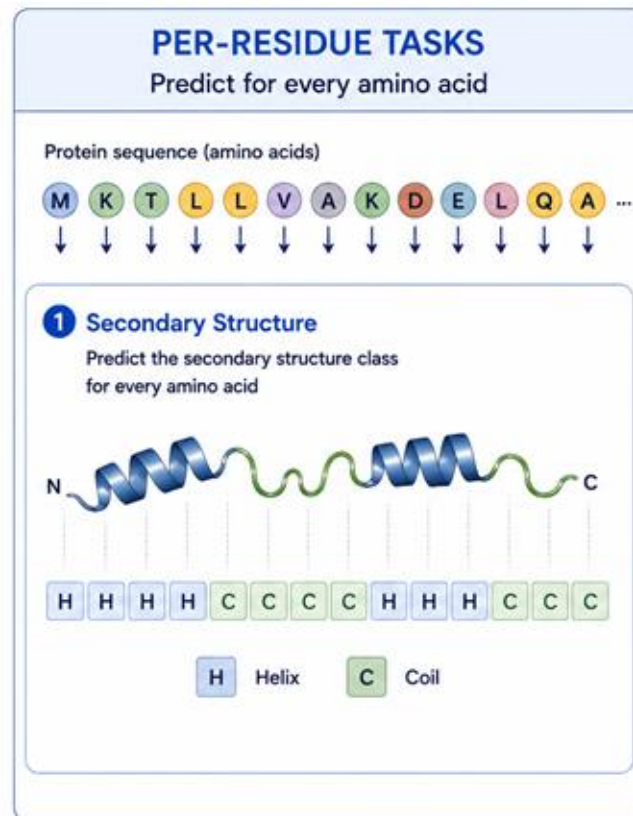
Per-Residue vs Per-Protein Tasks

Predict for every amino acid

- Secondary Structure
- Disorder

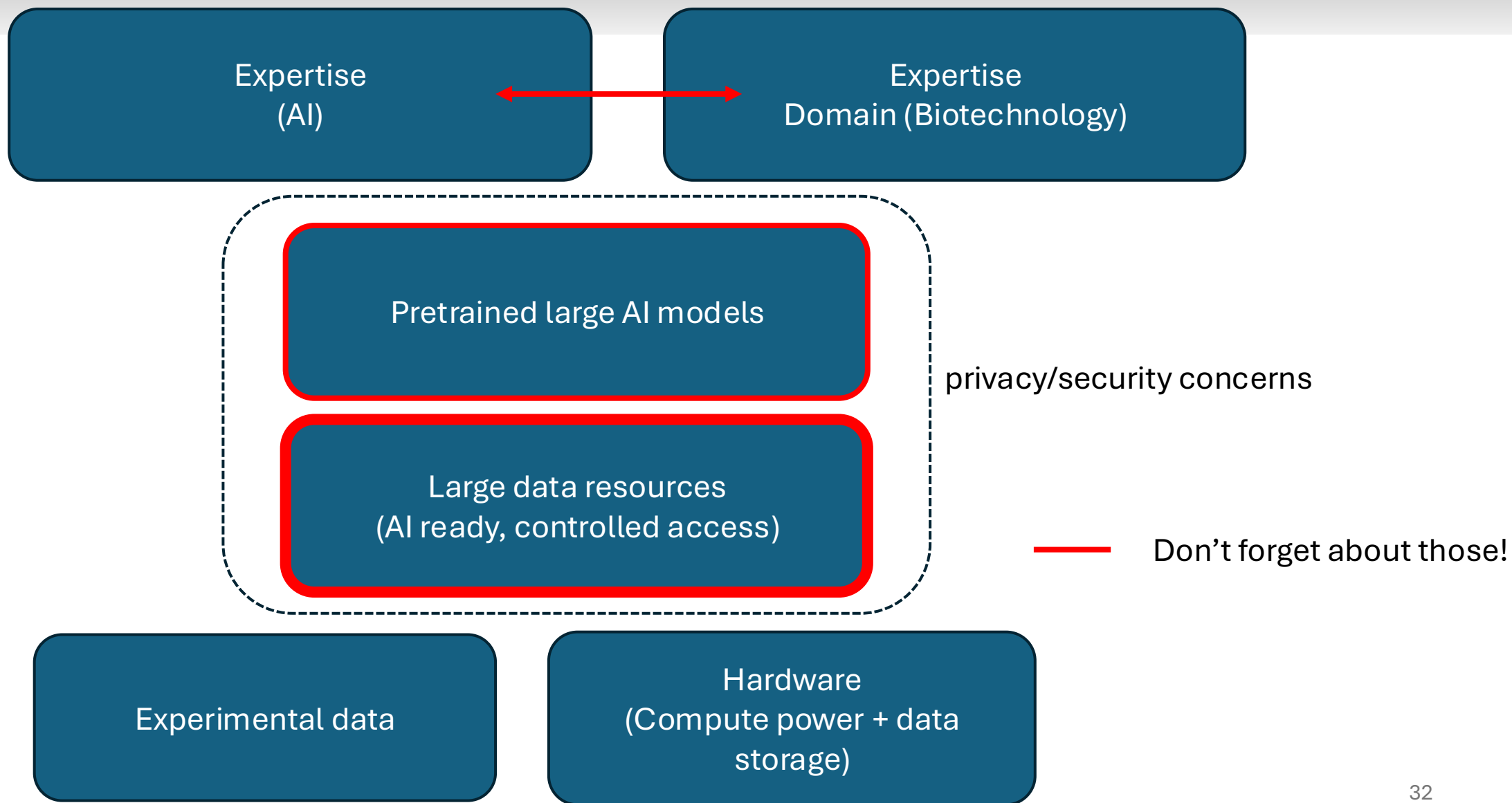
Predict one label for entire protein

- Localization
- Stability





Required AI infrastructure



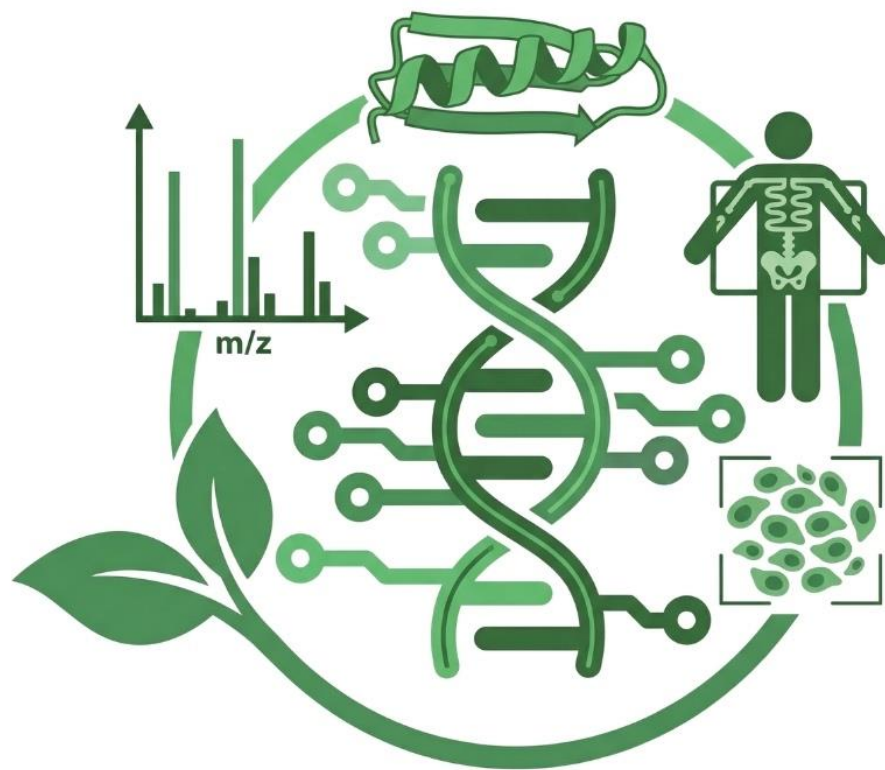


Key points

- Foundation models can be trained in a self-supervised manner
 - Help to learn representations
 - Can help to learn complex task, where little annotated data is available for end-to-end deep learning
- Protein language models, can help with a large number of downstream predictions tasks;
 - using either the embeddings, or by fine-tuning the model.



Advanced Machine Learning and AI for Life



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University